SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: BTech/MBATech/BTI | Year: II/IV Semester: IV/VIII

Stream/s ;Data Science/CSE DS(7057)

Subject: Introduction to Data, Signal and Image Analysis | Time: _03 hrs (10:00 am to 01:00 pm)

Date: 19 / 05 / 2025 | No. of Pages:__03___

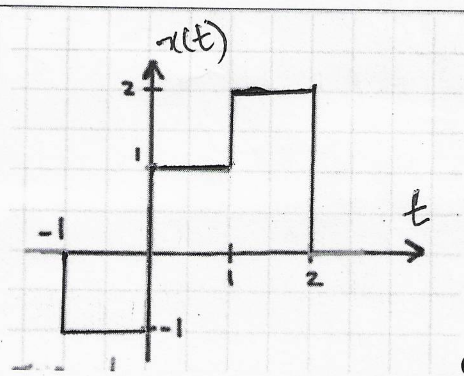
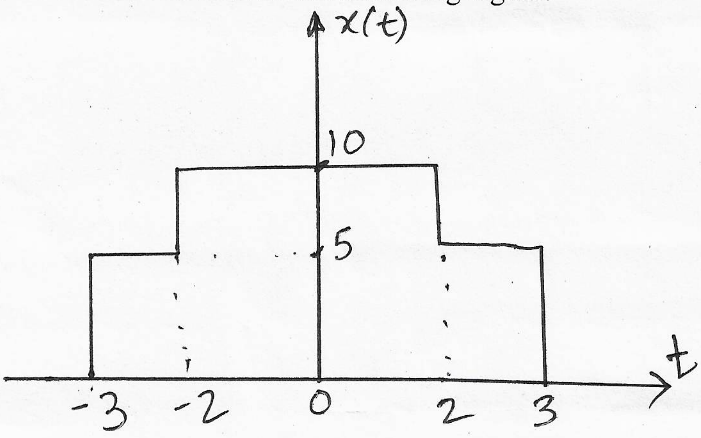
Marks: __100____

Final Examination 2024-25/Re-exam 2023-24

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1,BL-1,SO1	a.	Write the (equations/formula), of any 5 properties of Fourier Transform	[5]
CO-1, BL -3,SO2	b.	Solve to find x(0) and x(∞) for :- $$X(S) = \frac{5S+4}{s(2S+3)}$$	[5]
CO2,BL1,SO1	c.	List the steps in Impulse Invariant Method for IIR Filter design	[5]
Co1,BL1,SO1	d.	Write a note on continuous and discrete signals.	[5]
Q2 CO-1, BL-1; CO-3,BL-3,SO1	A	Explain the significance of sampling to convert continuous time into discrete. Apply sampling concept on a continuous time signal x(t)=20 sin(50πt) and find its discrete signal.	[10]

Q2 **CO1,BL3 ,SO1**	B	 Consider the signal x(t) given in the figure. Perform and sketch x $(t+1)$ and x $(-t/2 - 4)$	[10]
Q3 CO-3, BL-5,SO2	A	Determine Y(t) by using circular convolution. Given – $$x\ (t) = \ 1{,}3{,}2{,}1\ and\ h(t) = 1{,}1, -1{,}1$$	[10]
Q3 CO-1, BL-5,SO2	B	Evaluate Fourier Transform for the following signal: - 	[10]
Q4 **CO1,BL3 ,SO1**	A	Construct and sketch the signal X(t) = 2u(t+2) – u(t+1) – r(t-1) +r(t-2)	[10]
Q4 CO-1, BL-5,SO2	B	Evaluate- $L[\int_0^t t.e^{-7t} sin10t\ dt]$	[10]
Q5 CO-2, BL-2,SO1	A	List the steps for FIR Filter Design Using the Window Function	[10]

Q5 **CO4,BL2 ,SO1**	B	What is a Color Space? Why Are Color Spaces Important? Provide and explain examples of Popular Color Spaces	[10]
Q6 CO-2, BL-4,SO1	A	Compare Finite Impulse Response and Infinite Impulse Response filters in terms of the following aspects: 1. Stability 2. Computational complexity 3. Memory requirements 4. Phase response 5. Design complexity	[10]
Q6 CO4, BL4,SO1	B	1. "This Technique" is a more controlled process where the histogram of an image is adjusted to match a specific target histogram, often used for image standardization or matching appearance to a reference" 2. "This Method" enhances the contrast of an image by redistributing the intensity values across the full range, improving the visual quality and revealing more details" Which definition from the above two statements matches more closely with Histogram Equalization? Explain why?	[10]
Q7 CO4,BL4,SO1	A	Compare Mean vs Median filter.	[10]
Q7 CO-1, BL-1,BL3,SO1	B	Apply Fourier series (trigonometric form) for the given signal $x(t) = \begin{cases} 1 & \text{if } 0 \leq t < \pi \\ -1 & \text{if } \pi \leq t < 2\pi \end{cases}$	[10]